

Part 5: Properties of MLEs

Summary

In this Part we explore some of the important properties possessed by maximum likelihood estimators.

Briefly stated, we will see that, asymptotically, these estimators cannot be beat, in the sense that they have minimal MSE, and also are approximately normally distributed with variance that is easy to calculate. This makes quantifying the error in the estimators straightforward.

Invariance of the MLE

The invariance result can be stated as follows: If $\hat{\theta}_{\text{MLE}}$ is the MLE for θ , then $g(\hat{\theta}_{\text{MLE}})$ is the MLE for $g(\theta)$.

This is a useful, important result because it is often the case that we are truly interested in estimating some function of the parameter θ , not θ itself.

Using this, we can say, for example, that if $X_1, X_2, \dots, X_n$ are i.i.d. normal(μ, σ^2), then the MLE for σ is

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

and the MLE for the **signal to noise ratio** μ/σ is

$$\bar{X} / \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

One consequence of this property is that estimation is not affected by how one chooses to **parameterize** a model.

For example, some choose to represent the Exponential distribution using the “mean parameterization” which leads to $E(X) = \mu$, whereas we have used the “rate parameterization” which leads to $E(X) = 1/\lambda$. Because of invariance, these two approaches will lead to the same estimates, i.e., $\hat{\mu} = 1/\hat{\lambda}$.

Exercise: Does unbiasedness have this invariance property?

Exercise: Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. Exponential(λ). What is the MLE of the **tail probability** $P(X_i > a)$?

1 Consistency of the MLE

2 An estimator $\hat{\theta}$ is **consistent for θ** if as the sample size n increases, the estimator converges to the true value of the parameter.

4 Formally, we would write: For any $\epsilon > 0$,

$$5 \lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta_0| > \epsilon) = 0$$

6 This is a fairly low standard placed on an estimator, and, fortunately, under
7 certain “regularity conditions,” the MLE is consistent.

1 Asymptotic Normality of the MLE

2 Another great property of the MLE: The MLE is asymptotically normal,
3 with variance that is not difficult to calculate, under certain “regularity
4 conditions.” (See page 516 from Casella and Berger, Second Edition.)

5 This will allow an easy route to:

- 6 1. Calculate standard errors for MLEs
- 7 2. Constructing confidence intervals for parameters
- 8 3. Constructing confidence intervals for general functions of parameters
9 (via the Delta method)
- 10 4. Constructing hypothesis tests for parameters

1 To start, we focus on the case where the parameter θ is one-dimensional.

2 The formal result can be stated as follows:

3 If $X_1, X_2, \dots, X_n$ are i.i.d. $f(x; \theta_0)$, then, under suitable regularity conditions, as n increases,

$$5 \sqrt{n}(\hat{\theta} - \theta_0) \longrightarrow N(0, I^{-1}(\theta_0))$$

6 where

$$7 I(\theta_0) = E\left(-\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2}\right)$$

8 is the **Fisher Information**.

1 **Exercise:** Explain the practical value of the above result.

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1 **Exercise:** Show that

$$nI(\theta) = E \left[-\frac{\partial^2}{\partial \theta^2} \log L(\theta) \right]$$

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1 The previous result shows that the variance of the MLE is approximately
 2 equal to the reciprocal of the expected value of the negative second deriva-
 3 tive of the log likelihood at its peak.

4 And, “the negative of the second derivative” is the amount of curvature at
 5 the peak of the likelihood.

6 If the log likelihood has a “sharp” peak, then $nI(\theta)$ will be large, and the
 7 variance of $\hat{\theta}$ will be small.

1 **Exercise:** Assume $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, where σ^2 is known. We
 2 know that $\bar{X}$ is the MLE for μ . Derive the Fisher Information for μ , and use
 3 it to approximate the standard error and distribution of $\bar{X}$.

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- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. Derive the
2 Fisher Information for λ and the approximate distribution of its MLE.

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- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Poisson}(\lambda)$. Derive the
2 Fisher Information for λ and the approximate distribution of its MLE.

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1 Efficiency of the MLE

- 2 It is possible to show that the lower bound on the variance of any unbiased
3 estimator is $1/nI(\theta_0)$. This is called the **Cramer-Rao lower bound**.
4 Since the MLE is asymptotically unbiased, we know that we are achieving
5 the Cramer-Rao lower bound, at least asymptotically.
6 Hence, we are minimizing the MSE (asymptotically). This is objectively
7 the best rationale for utilizing maximum likelihood estimators.

1 The Delta Method

- 2 Often, our objective is not to estimate θ , but instead some function of θ , call
3 it $g(\theta)$. For example, often want to estimate the probability of some event
4 A . This probability is not a parameter, but it is a function of θ .
5 From the invariance property we know that the MLE of $g(\theta)$ is $g(\hat{\theta})$. But,
6 we also need to calculate a SE or construct a confidence interval for this
7 estimate.
8 One approach would be to “reparameterize” the model in terms of $g(\theta)$,
9 and then find $I(g(\theta))$, and then use $1/I(g(\theta))$ as the variance, etc. Fortu-
10 nately, there is an easier way...

1 Recall the **Delta Method**:

2 If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \rightarrow 0$ as n increases, then
 3 $g(Y_n)$ is approximately $N(g(\mu), (g'(\mu))^2 \sigma_n^2)$, assuming that $g'(\mu)$ ex-
 4 ists and is not zero.

5 Here we will use this result with the MLE for θ in place of Y_n .

6 Then, $\mu = \theta_0$ and $\sigma_n^2 = 1/nI(\theta_0)$.

7 We can conclude that $g(\hat{\theta})$ is approximately

$$8 \quad N\left(g(\theta_0), g'(\theta_0)^2 \left(\frac{1}{nI(\theta_0)}\right)\right)$$

9 In practice, we use the approximate variance:

$$10 \quad g'(\hat{\theta})^2 \left(\frac{1}{nI(\hat{\theta})}\right)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. We seek to
 2 estimate $\tau = P(X_i > a)$ for some constant $a > 0$. Construct the MLE and
 3 find its standard error. Also construct a confidence interval for τ .

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1 **The Multidimensional Case**

2 Now we consider the case where $\theta = (\theta_1, \theta_2, \dots, \theta_p)$.

3 We define the **Fisher Information matrix** to be the p by p matrix whose (i, j)
 4 entry is

$$5 \quad I_{ij}(\theta) = E\left[-\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(x; \theta)\right]$$

6 **The Main Result:** Under appropriate “regularity conditions,” if $X_1, X_2, \dots, X_n$
 7 are i.i.d. $f(x; \theta)$ then

$$8 \quad \sqrt{n}(\hat{\theta} - \theta_0) \longrightarrow N_p(0, \mathbf{I}^{-1}(\theta_0))$$

9 Recall that N_p denotes the p -dimensional multivariate normal distribution.

10 **Practical Version:** Under “regularity conditions,” $\hat{\theta}$ is approximately nor-
 11 mal with variance $\mathbf{I}^{-1}(\hat{\theta})/n$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$. Derive the joint
 2 distribution of the MLE for $\theta = (\mu, \sigma^2)$.

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1 Multivariate Delta Method

2 We can extend the Delta method to cases where the parameter is multidi-
 3 mensional.

4 In general, suppose that $\mathbf{Y}_n$ is approximately $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}/n)$. Now consider
 5 an m -dimensional function $\mathbf{g}(\cdot)$ where $m \leq p$:

$$\mathbf{g}(\mathbf{Y}_n) = \begin{bmatrix} g_1(\mathbf{Y}_n) \\ g_2(\mathbf{Y}_n) \\ \vdots \\ g_m(\mathbf{Y}_n) \end{bmatrix}$$

7 Then, $\mathbf{g}(\mathbf{Y}_n)$ is approximately $N_m(\mathbf{g}(\boldsymbol{\mu}), \mathbf{G}\boldsymbol{\Sigma}\mathbf{G}^T/n)$ where the (i, j) entry of
 8 $\mathbf{G}$ is $\partial g_i(\mathbf{y})/\partial y_j$. This all assumes that $\mathbf{G}$ is of full rank.

- 1 **Exercise:** Place our parameter estimation problem into the framework of
 2 the above statement of the Delta method.

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1 **Case Study: Geometric Brownian Motion**

- 2 This Part concludes with a simple, but important, example of how careful
 3 derivation of the MLE and its distribution can lead to useful insight.
- 4 We begin by defining some standard concepts regarding Brownian motion.

1 *The Standard Brownian Motion (Wiener Process)*

- 2 A stochastic process $\{W(t) : t \geq 0\}$ is a **(standard) Brownian motion** if the
 3 following hold:
- 4 1. $W(0) = 0$. (almost surely)
 - 5 2. $W(t)$ is continuous as a function of t . (almost surely)
 - 6 3. $W(t_4) - W(t_3)$ and $W(t_2) - W(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
 - 7 4. The distribution of $W(t_2) - W(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
 - 8 5. $W(t)$ is Normal($0, t$) for all t .

1 *Brownian Motion with Drift and Scaling*

- 2 A stochastic process $\{B(t) : t \geq 0\}$ is a **Brownian Motion with drift μ and**
 3 **scaling σ^2** if $\{W(t) : t \geq 0\}$ is a standard Brownian motion and

$$4 \quad B(t) = \mu t + \sigma W(t).$$

- 5 Note that this implies the following:

- 6 1. $B(0) = 0$. (almost surely)
- 7 2. $B(t)$ is continuous as a function of t . (almost surely)
- 8 3. $B(t_4) - B(t_3)$ and $B(t_2) - B(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 4. The distribution of $B(t_2) - B(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 10 5. $B(t)$ is Normal($\mu t, \sigma^2 t$) for all t .

1 Geometric Brownian Motion

2 A stochastic process $\{S(t) : t \geq 0\}$ is a **geometric Brownian motion** if $\{B(t) :$
 3 $t \geq 0\}$ is a Brownian motion with drift μ and scaling σ^2 and

$$4 \quad S(t) = S(0) e^{B(t)}$$

5 where $S(0) > 0$ is the initial value. Define $\alpha = \mu + \sigma^2/2$.

6 Note that this implies the following:

- 7 1. $S(t)$ is continuous as a function of t . (almost surely)
- 8 2. $S(t_4)/S(t_3)$ and $S(t_2)/S(t_1)$ are independent for any $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 3. The distribution of $S(t_2)/S(t_1)$ depends only on $|t_1 - t_2|$ for any t_1, t_2 .
- 10 4. $S(t)$ is Lognormal($\log(S(0)) + \mu t, \sigma^2 t$) for all t .
- 11 5. $S(t)$ has expected value $S(0)e^{\alpha t}$.
- 12 6. $\log(S(t_2)/S(t_1))$ is Normal($\mu(t_2 - t_1), \sigma^2(t_2 - t_1)$) for all $t_1 \leq t_2$.

1 Let's consider the geometric Brownian motion model. Can we use data
 2 (observed prices) to estimate α and σ^2 ?

3 Suppose we observe the stock price over the period $[0, T]$. We will model
 4 the price as a geometric Brownian motion $\{S(t) : t \geq 0\}$.

5 Divide the time interval into n equal-width subintervals, with $\Delta = T/n$.

6 Define, for $i = 1, 2, \dots, n$,

$$7 \quad Y_i = \log \left(\frac{S(t_{i-1} + \Delta)}{S(t_{i-1})} \right)$$

8 where $t_i = i/n$, the times at which the price is observed.

9 From Property 6 of a geometric Brownian motion, $Y_1, Y_2, \dots, Y_n$ are i.i.d.

10 $N(\Delta\mu, \Delta\sigma^2)$.

1 **Exercise:** Find the MLE of $\theta = (\alpha, \sigma^2)$. Approximate the distribution of θ .

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Exercise: What does this result suggest regarding how large n should be chosen for the best estimation of σ^2 ? What about for α ?

Is it, in fact, a good idea to sample more frequently in order to estimate σ^2 ? See Ait-Sahalia, Mykland, and Zhang (2005), "How Often to Sample a Continuous-Time Process in the Presence of Market Microstructure Noise," *The Review of Financial Studies*, Volume 18.